

Logarithmic laws



- 1
- | | | | |
|---|--------------------|--|-------------------|
| a $\log_{10} 20$ | b $\log_{10} 6$ | c $\log_{10} \left(\frac{1}{4} \right)$ | d $\log_5 198$ |
| e $\log_{10} \left(\frac{35}{3} \right)$ | f $\log_7 2$ | g $3 \log_{10} 18$ | h $3 \log_{10} 3$ |
| i $2 \log_5 2$ | j $5 \log_{10} 18$ | k $\log_4 448$ | l $\log_3 (10xy)$ |

- 2
- | | | | |
|------|-----|-----|-----|
| a 1 | b 3 | c 3 | d 2 |
| e -2 | f 2 | | |

- 3
- | | | | |
|----------------|--------------------------|--------------------------|------------------------|
| a $7 \log_4 x$ | b $5 \log (x + 6)$ | c $-\log (3x + 7)$ | d $\frac{2}{5} \log x$ |
| e $5 \log_6 y$ | f $\frac{1}{2} \log_6 w$ | g $\frac{1}{2}$ | h $\frac{5}{4}$ |
| i a | j 10 | k $\frac{3}{2} \log_5 x$ | l $\frac{15}{4}$ |

- 4
- | | |
|------------------------------------|-----------------------------------|
| a $\log_9 u + \log_9 v$ | b $\log_5 9 - \log_5 7$ |
| c $\log_2 5 + \log_2 x$ | d $\log_{10} 2 - \log_{10} x$ |
| e $\log m + \log m$ | f $5 (\log 3 + \log x)$ |
| g $\log p + \log q - \log r$ | h $-\log x - \log y$ |
| i $-7 (\log 5 + \log x)$ | j $\log 5 + \frac{2}{3} \log x$ |
| k $\frac{1}{3} (\log 14 + \log x)$ | l $4 \log c - \frac{1}{2} \log d$ |

- 5
- | | | | |
|---------------------|--------------------|---|---------------------------------------|
| a $7 \log x$ | b $\log (x^5 y^3)$ | c $\log \left(\frac{x^8}{y^{\frac{1}{3}}} \right)$ | d $\log \left(\frac{x^8}{y} \right)$ |
| e $-35 \log_{10} 5$ | f $9 \log_{10} 2$ | g 2 | h 2 |

6 $\log 2 + \log u - \log 3 - \log v$

7 $5 \log x$

8

a $3 \log u + 5 \log v$	b $\frac{1}{3} \log v - \frac{1}{2} \log u$
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9

a 5

b 2



c 9

d -4

e 2

f 3

g 5

h 2

10 2.105

11

a 1.447

b 0.222

c 3.121

d -0.407

12

a 1.24

b 0.31

c 1.78

d -0.62

e 2.02

13

a 7.8

b 1.4

c 7.3

d -1.6

14 $\frac{\log_4 20}{\log_4 3}$

15

a $\frac{\log_{10} 16}{\log_{10} 4}$

b $\frac{\log_{10} 0.9}{\log_{10} 3}$

c $\frac{\log_{10} \sqrt{5}}{\log_{10} 3}$

16

a i $\frac{\log_{10} 21}{\log_{10} 8}$

ii 1.46

b i $\frac{\log_{10} \sqrt{5}}{\log_{10} 4}$

ii 0.58

c i $\frac{\log_{10} 3^3}{\log_{10} 2}$

ii 4.75

d i $\frac{\log_{10} 105}{\log_{10} \pi}$

ii 4.07

17

a $p + q$

b $p - q$

c $2q$

d $2p + 1$